

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20							***** MODULAR VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
30										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30			MODULAR DBB	K01AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0		
40	U04		MODULAR DBB	M01AC02	0.75	2	MODULAR VALVE DB&B BALL TYPE 150# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO	0	0		
50	U04		MODULAR DBB	M01AM14	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR-HI-TEMP	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
60	U04		MODULAR DBB	K01AM14	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV-HI-TEMP	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
70		VAV	MODULAR DBB	K03AM10	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0	ND8	USE ONLY WHERE SPACE IS LIMITED AND WITH APPROVAL BY MATERIALS ENGINEER.
80			MODULAR DBB	M01AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	UG3	PTFE GASKET ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE)
90		FS1	MODULAR DBB	M03AM10	0.75	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	UG4	BOLTS ONLY TO BE USED FOR INTERFACE WITH POLYETHYLENE (HDPE) UNDERGROUND PIPING
100		VAW	MODULAR DBB	M03EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2		
110	U04	FS1	MODULAR DBB	M03AM14	0.75	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR-HI-TEMP	0	0		
120	U04	VAV	MODULAR DBB	K03AM14	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV-HI-TEMP	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
130		VAW	MODULAR DBB	M03EM14	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR-HI-TEMP	2	3	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
140		VAW	MODULAR DBB	M03EM14L	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR-HI-TEMP	2	3	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
150		VB2					***** BALL VALVES *****			VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
160	U04	VB2	BALL	A01AC01	3	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB LO	0	0		
170	U04		BALL	A01AC02	3	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB LO	0	0		
180	U04	VB2	BALL	A01AC03	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT RB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
190	U04		BALL	A01AC04	6	6	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL RB GO	0	0		
200	U04	VB2	BALL	A01AC07	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
210	U04		BALL	A01AC08	8	8	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0		
220	U04	VB2	BALL	B01AC01	0.75	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB LO	0	0		
230			BALL	A01AM09	3	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0		
240	U04		BALL	A01AM10	2	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING METAL RB LO	0	0		
250	U04		BALL	A01AR09	4	4	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0		
260	U04	VA5	BALL	B01AC02	0.75	3	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB LO	0	0		
270	U04		BALL	A01AR15	6	16	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT RB GO	0	0		
280		VA5	BALL	B01AM09	0.5	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0		
290		VA5	BALL	B01AR11	4	4	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT FB GO	0	0		
300		VA5	BALL	B01AR15	6	24	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT FB GO	0	0		
310	U04	VA5	BALL	B01AC03	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING SOFT FB GO	0	0		
320	U04	VA5	BALL	B01AC04	4	4	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM FLOATING METAL FB GO	0	0		
330	U04	VA5	BALL	B01JC08	26	60	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO B16.47 SERIES A	0	0		
340		FS1	BALL	B03AM09	0.75	2	BALL 300# RF INC 825 INC 625 TRIM FLOATING SOFT FB LO	0	0		
350	U04	VA5	BALL	B01AC07	6	24	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
360	U04	VA5	BALL	B01AC08	6	24	BALL 150# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
370		FS1	BALL	A03AM09	3	3	BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0		
380							***** BUTTERFLY VALVES *****				
390			BUTTERFLY	T01AC61	10	24	BUTTERFLY 150# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
400			BUTTERFLY	T01AR63	10	24	BUTTERFLY 150# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO	0	0		
410							***** GATE VALVES *****				
420	U04		GATE	G01AM49	0.5	3	GATE 150# RF INC 825 INC TRIM 625 WEDGE HW	0	0		
430			GATE	G01AR43	4	8	GATE 150# RF LTCS + CLAD INC 825 INC TRIM 625 WEDGE HW	0	0		
440		VAV	GATE	G03AM49	0.75	2	GATE 300# RF INC 825 INC 625 TRIM WEDGE HW	0	0		
450							***** GLOBE VALVES *****				
460							*****BUTTERFLY VALVE*****				
470											
470			GLOBE	L01AM49	0.75	3	GLOBE 150# RF INC 825 INC TRIM 625 OS&Y HW	0	0		
480			GLOBE	L01AR49	4	6	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW	0	0		
490			GLOBE	L01AR44	8	10	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y GO	0	0		
500							***** CHECK VALVES *****				
510											
510		VC1	CHECK	C01AM51	0.75	1.5	CHECK PISTON 150# RF INC 825 INC TRIM 625	0	0		
520			CHECK	C01AM46	2	3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED	0	0		

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR WT > 19 MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS RATING: 150 RF TEMPERATURE RANGE: -40 /215 °C	12/16/2024	BY OZAC	U04	Page 1 of 3	150H23

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530			CHECK	C01AR46	4	6	CHECK DUAL WAFER LUG 150# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED	0	0	
540			CHECK	C01AR53	8	16	CHECK DUAL DBL FLG 150# RF LTCS + CLAD INC 825 INC TRIM 625	0	0	
550							***** PIPE *****			
550										
560										
560			PIPE	5358511	0.5	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0	
570			PIPE	5358513	2	2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0	
580			PIPE	5358514	3	16	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0	
590			PIPE	5358577	18	24	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE	0	0	
600							***** FITTINGS *****			
600										
610										
610			ELBOW	5392870	0.75	2	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0	
620			ELBOW	5392600	3	16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0	
630			ELBOW	5392497	18	24	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6-W	0	0	
640	U04	ND8	ELBOW SR	5392246	20	20	ELL BW 90 DEG SR SCH XS LTCS A420 WPL6-W	0	0	
650			ELBOW	5392876	0.75	2	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0	
660			ELBOW	5392606	3	16	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0	
670			ELBOW	5392608	18	24	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6-W	0	0	
680										
680			TEE	5392879	0.75	2	TEE SCH XXS LTCS A420 WPL6	0	0	
690			TEE	5392609	3	16	TEE BW SCH XS LTCS A420 WPL6	0	0	
700			TEE	5392406	18	24	TEE BW SCH XS LTCS A420 WPL6-W	0	0	
710			TEE RED	5392877	0.75	1.5	TEE RED SCH XXS LTCS A420 WPL6	1	2	
720			TEE RED	5392897	1.5	2	TEE RED BW SCH XS/XXS LTCS A420 WPL6	3	4	
730			TEE RED	5392661	3	14	TEE RED BW SCH XS LTCS A420 WPL6	4	16	
740			TEE RED	5392413	8	20	TEE RED SCH XS LTCS A420 WPL6-W	18	24	
750										
750	U04		CAP	5392880	0.75	2	CAP BW SCH XXS LTCS A420 WPL6	0	0	
760			CAP	5392610	3	24	CAP BW SCH XS LTCS A420 WPL6	0	0	
770										
770			RED CONC	5392881	0.75	1.5	REDUCER CONC SCH XXS LTCS A420 WPL6	1	2	
780			RED CONC	5392898	1.5	2	REDUCER CONC SCH XS/XXS LTCS A420 WPL6	3	4	
790			RED CONC	5392458	3	14	REDUCER CONC SCH XS LTCS A420 WPL6	4	16	
800			RED CONC	5392507	10	20	REDUCER CONC SCH XS LTCS A420 WPL6-W	18	24	
810			RED ECC	5392882	0.75	1.5	REDUCER ECC SCH XXS LTCS A420 WPL6	1	2	
820			RED ECC	5392899	1.5	2	REDUCER ECC SCH XS/XXS LTCS A420 WPL6	3	4	
830			RED ECC	5392461	3	14	REDUCER ECC SCH XS LTCS A420 WPL6	4	16	
840			RED ECC	5392514	10	20	REDUCER ECC SCH XS LTCS A420 WPL6-W	18	24	
850										
850	U04		OLETS	5504211	0.75	1.5	NIPOFLANGE 150# RF SCH XXS LTCS A350 LF2 CL.1	2	2	
860	U04		OLETS	5504212	0.75	2	NIPOFLANGE 150# RF SCH XS/XXS LTCS A350 LF2 CL.1	3	24	
870	U04		OLETS	5504215	1	1.5	NIPOFLANGE 300# RF SCH XXS LTCS A350 LF2 CL.1	1.5	2	
880	U04	UJ2	OLETS	5504216	0.75	1.5	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	3	6	
890	U04	UJ2	OLETS	5504213	1.5	1.5	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	4	16	
900										
900	U04		OLETS	5505197	0.75	1	WELDOLET SCH XXS LTCS A350 LF2 CL.1	2	2	
910	U04		OLETS	5505180	0.75	2	WELDOLET SCH XS/XXS LTCS A350 LF2 CL.1	3	24	
920	U04	UJ2	OLETS	5505335	3	6	WELDOLET SCH XS LTCS A350 LF2 CL.1	8	24	
930										
930	U04	UJ2	ELBOFLG	5516762	1.5	1.5	ELBOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	4	16	
940										
940		FS1					***** FLANGES *****			
950										
950	U04	FA8	FLANGE	5599692	0.5	0.5	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1	0.75	1.5	
960	U04	FA8	FLANGE	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	0.75	
970										
970	U04		FLG ADAPTER	5851200	0.5	2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0	
980	U04		FLG ADAPTER	5851203	0.5	2	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
990										
990	U04		FLANGE	5556221	0.5	2	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
1000	U04		FLANGE	5556264	3	24	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1010	U04		FLANGE	5556481	0.75	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
1020	U04		FLANGE	5556476	3	16	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1030	U04		BLIND	5615566	0.75	24	FLG BLIND 150# RF LTCS A350 LF2 CL.1	0	0	
1040	U04		BLIND	5615576	0.75	16	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
1050	U04		FLANGE	5556276	6	6	FLG WN REDUCING 150# RF LTCS A350 LF2 CL.1 SCH XS	10	10	
1060							***** ORIFICE FLANGES *****			
1060										
1070										
1070	U04		ORIFICE FLG	5623947	2	24	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
1080							***** PIPE NIPPLES *****			
1080										

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1090										
1090		812		0			NIPPLE PBE - CUT FROM PIPE			
1100										
1100		FS1					***** GASKETS *****			
1110										
1110	U04		GASKET	5673043	0.5	24	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1120			GASKET	5673033	0.75	16	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1130	U04	UG3	GASKET	5664201	10	10	GASKET 150# RF FLAT RING PTFE	0	0	
1140										
1140		B1					***** BOLTING *****			
1150										
1150	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1160	U04	UG4		5675768	10	10	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) W/ 2 NUTS + LTCS WASHERS	0	0	
1170										
1170							***** BLINDS & SPACERS *****			
1180										
1180			SPEC BLIND	5749580	0.75	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70	0	0	
1190			SPEC BLIND	5749581	1.5	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
1200										
1200			PADDLE	5750023	16	24	PADDLE BLANK 150# RF LTCS A516 GR.70	0	0	
1210			PADDLE	5750100	12	16	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
1220	U04		PADDLE	5750240	16	24	PADDLE SPACER 150# RF LTCS A516 GR.70	0	0	
1230	U04		PADDLE	5750200	12	16	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1240										

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